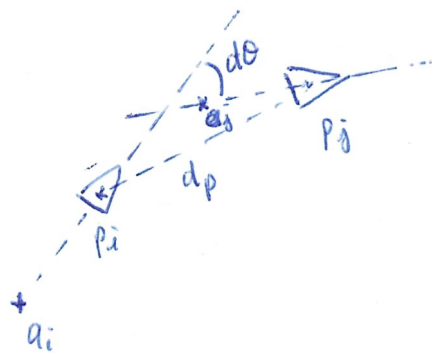


Ex-29. Consensus



If we consider the anchor a_j at a meters of p_j , we have in R_j :

$$\vec{a_j} = \vec{p_j} - a_j \begin{pmatrix} \cos(\theta_j - \theta_i) \\ \sin(\theta_j - \theta_i) \end{pmatrix}$$

$\vec{a_j}$ in R_j

If R_i follows the Robot $R_j = R_{i+1}$, we have the relative position d_{pi}

$$d_{pi} = \Delta(p_{ji}) = R(\theta_i) \cdot (p_j - p_i)$$

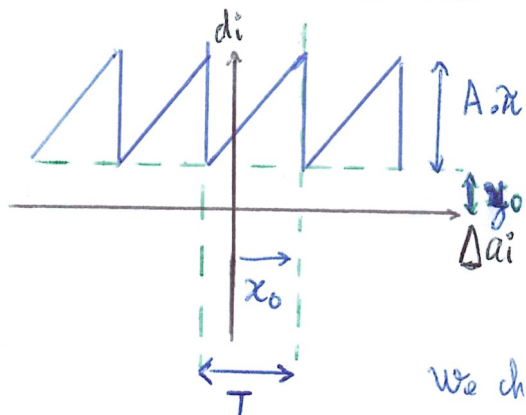
$$\text{so } w_i = d_{pi} - a_j \begin{pmatrix} \cos d\theta \\ \sin d\theta \end{pmatrix} + \underbrace{\sum (\text{repulsive force})}_{-\sum_{i \neq j} \frac{d_{pi}}{\|d_{pi}\|^3}}$$

We choose

$$\begin{aligned} u_1 &= \text{Sawtooth} \left(\text{atan} \left(\frac{w_y}{w_x} \right) \right) \\ u_2 &= k_x (\|w\| - v) \end{aligned}$$

Ex-30. Platooning.

1) we have $x(i) = (a_i, v_i)$ and we want to find $g(a_i, a_i, v_i, v_i)$ that give us $y(i) = (d_i, d_i)$, where d_i is the circular distance determined by $\Delta a_i = a_i - a_i$ and modulo L . We will modify the sawtooth function to get that distance:



$\sigma = \text{Sawtooth image}$ is in $[-\pi, \pi]$ with a period of 2π .

We want σ_d with period L , ~~between~~ and image in $[0, L]$.

$$\text{We have } \sigma_d(x) = A \cdot \left(\underbrace{\sigma \left(\frac{2\pi}{T} (x - x_0) \right)}_{\in [-\pi, \pi]} + y_0 \right)$$

$$\text{We choose } \begin{cases} x_0 = L/2 \\ y_0 = \pi \\ T = L = 2\pi \cdot n \\ A = n \end{cases}$$

$$\begin{aligned} &\in [0, 2\pi] \text{ if } y_0 = \pi \\ &\in [0, L] \text{ if } A = n \end{aligned}$$

$$So \quad g(a_{i-}, a_i, v_{i-}, v_i) = \begin{bmatrix} \sigma_d(a_{i-} - a_i) \\ v_{i-} - v_i \end{bmatrix}$$

2) We choose $u_i = \underbrace{k_1(d_i - d_0)}_{\text{control of distance}} + \underbrace{k_2 \dot{d}_i}_{\text{speed damping}} + \underbrace{k_3(v_0 - v_i)}_{\text{speed control}}$ where $k_1, k_2, k_3 \in \mathbb{R}^{+*}$

3) We call $p = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_{m-1} \\ v_1 \\ \vdots \\ v_m \end{pmatrix}$. In case of stability, $\dot{p} = \vec{0}$

$$\dot{p} = \vec{0} \Leftrightarrow \begin{cases} \forall i \in [1, m], \quad \dot{d}_i = 0 \\ \quad \quad \quad \dot{v}_i = 0 \end{cases} \Leftrightarrow \begin{cases} v_{i-} - v_i = 0 \\ u_i = 0 \end{cases} \Leftrightarrow \begin{cases} v_{i-} = v_i \\ d_i - d_0 + \dot{d}_i + v_0 - v_i = 0 \end{cases}$$

$k_1 = k_2 = k_3 = 1$

$$\dot{p} = \vec{0} \Leftrightarrow \forall i \in [1, m], \quad \begin{cases} v_{i-} = v_i \\ v_i = d_0 - d_i + v_0 \end{cases}$$

$$\Leftrightarrow \forall i \in [1, m], \quad \begin{cases} v_{i-} = v_i \\ d_0 - d_{i-} + v_0 = d_0 - d_i + v_0 \end{cases}$$

$$\Leftrightarrow \forall i \in [1, m], \quad \begin{cases} v_{i-} = v_i \\ d_{i-} = d_i \end{cases} \quad \text{but } \begin{cases} L = \sum_{i=1}^m d_i \\ L = m \cdot d_0 \end{cases} \Rightarrow \begin{cases} L = m \times d_i \\ L = m \times d_0 \end{cases} \Rightarrow \boxed{d_i = d_0}$$

$$\text{Then } d_i - d_0 + v_0 - v_i = 0 \Rightarrow \boxed{v_i = v_0} \quad d_i = d_0$$

4) With 4 robots; applying $u_i = \dot{v}_i \Rightarrow \begin{cases} \dot{d}_i = v_{i-1} - v_i \\ \dot{v}_i = d_i - d_0 + \dot{d}_i + v_0 - v_i \end{cases}$
we then can have:

$$\dot{p} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ -1 & 0 & 0 & -2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -2 & 0 \\ -1 & -1 & -1 & 0 & 0 & 1 & -2 \end{pmatrix}}_A p + \begin{pmatrix} v_0 - d_0 \\ \vdots \\ v_0 - d_0 \\ v_0 - d_0 + L \end{pmatrix} \quad \text{with } p = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

we can show that the eigenvalues of A are strictly negatives, so the system is stable.